Smart Water Logging, Flood Prevention and Post Flood Event Management System

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Project Proposal Report

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DECLARATION

DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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ABSTRACT

Floods are among the most frequent and destructive natural disasters, particularly in countries such as Sri Lanka, where seasonal monsoon rainfall and inadequate drainage systems often lead to severe flooding. While significant advancements have been made in flood prediction and monitoring, existing systems largely focus on pre-disaster analysis and lack effective mechanisms for post-flood response. In particular, there is a critical gap in prioritizing affected locations, allocating appropriate relief resources, and optimizing delivery operations in real time.

To address these limitations, this research proposes an intelligent post-flood management system that integrates real-time data processing, machine learning, and geospatial analysis to support efficient disaster response and recovery. The system identifies safe zones, detects refugee camps, and analyzes camp-specific conditions such as population size, resource availability, and health risks. A supervised machine learning model, specifically XGBoost, is employed to classify and prioritize refugee camps based on urgency. In addition, a rule-based mechanism is used to determine the priority of essential relief items, including food, water, medicine, and sanitary supplies. Furthermore, route optimization algorithms such as Dijkstra and A* are applied to identify the safest and most efficient paths for aid distribution.

The system is developed using technologies such as React.js for the frontend, Node.js for backend processing, MySQL for data management, and Python-based libraries for machine learning implementation. The expected outcomes include improved prioritization accuracy, reduced response time, efficient resource allocation, and enhanced coordination among stakeholders.

From a social perspective, the system contributes to saving lives, reducing disaster impact, and improving resilience in vulnerable communities. Commercially, it offers potential as a scalable disaster management solution for government agencies and humanitarian organizations through Software-as-a-Service (SaaS) and licensing models.

Keywords: Flood Management, Machine Learning, Disaster Response, Resource Optimization, Geospatial Analysis

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LIST OF ABBREVIATIONS

Abbreviation	**Full Form**
ML	Machine Learning
RF	Random Forest
DFD	Data Flow Diagram
GIS	Geographic Information System
UI	User Interface
API	Application Programming Interface
IoT	Internet of Things
GPS	Global Positioning System
SDG	Sustainable Development Goals
DM	Disaster Management
IT	Information Technology
DB	Database
SQL	Structured Query Language
KPI	Key Performance Indicator

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1. INTRODUCTION

1.1 Background and Context

Floods are among the most frequent and destructive natural disasters worldwide, causing extensive damage to infrastructure, displacement of communities, and significant socio-economic disruption. In countries such as Sri Lanka, seasonal monsoon rainfall, combined with inadequate drainage systems and rapid urbanization, frequently results in severe flooding in both urban and rural areas. While considerable advancements have been made in flood prediction and monitoring systems, a critical challenge remains in effectively managing post-flood response activities, particularly in coordinating rescue operations and distributing essential resources to affected populations.

Flood disasters disproportionately affect vulnerable groups, including low-income communities, children, the elderly, and individuals with pre-existing health conditions. Displaced populations residing in temporary refugee camps often face shortages of basic necessities such as clean water, food, clothing, and medical supplies. Furthermore, unsanitary conditions and overcrowding in such camps increase the risk of disease outbreaks. The timely and efficient distribution of relief resources is therefore crucial, as delays or mismanagement can lead to severe health consequences, increased mortality, and prolonged recovery periods.

Statistical evidence highlights the global and local significance of this issue. Floods account for a substantial proportion of natural disasters worldwide, affecting millions of people annually and resulting in considerable economic losses. In Sri Lanka, flood events are recurrent during monsoon seasons, with thousands of families displaced each year and critical infrastructure frequently disrupted. These statistics emphasize the urgent need for improved systems that not only predict flood occurrences but also enhance post-disaster response and recovery processes.

Existing solutions in flood management primarily focus on early warning systems, water level monitoring using IoT sensors, and flood risk mapping. Although these systems provide valuable insights for disaster preparedness, they exhibit several limitations in post-flood scenarios. Most notably, there is a lack of integration between flood detection and relief distribution, limited real-time decision-making capabilities, absence of prioritization mechanisms for affected locations, and inefficient allocation of resources. Additionally, current systems rarely incorporate item-level prioritization or safe route planning for aid delivery, resulting in delayed and uncoordinated response efforts.

To address these limitations, this research proposes an intelligent, integrated post-flood management system that combines real-time data collection, machine learning, and decision-support mechanisms. The proposed solution includes modules for flood map generation, drainage monitoring and prediction, disease detection, and a relief prioritization and distribution component. Specifically, the system identifies safe zones using flood data, locates refugee camps within these zones, and analyzes camp-specific conditions such as population size, resource availability, and health risks. A machine learning model is then employed to predict the priority level of each camp, while a rule-based mechanism determines the priority of required relief items, including food, water, medicine, and essential supplies. Furthermore, the system incorporates route optimization techniques to identify the safest and most efficient paths for aid delivery, thereby enhancing coordination and response efficiency.

This research aligns with key Sustainable Development Goals (SDGs), particularly SDG 11 (Sustainable Cities and Communities) and SDG 13 (Climate Action), by contributing to disaster resilience and improved emergency response systems. Moreover, the study falls within the Smart Cities and Digital Transformation research cluster, as it leverages Internet of Things (IoT) technologies and artificial intelligence to develop a data-driven, scalable, and efficient solution for disaster management and recovery.

1.2 Literature Review

Machine learning techniques have increasingly been applied in disaster management to improve decision-making and emergency response efficiency. Li et al. proposed a deep reinforcement learning–based rescue path planning system for urban flood scenarios. The model evaluates environmental conditions and dynamically identifies safer routes for rescue teams. This approach improves navigation efficiency during disasters and reduces response time. However, the study mainly focuses on route optimization rather than identifying which affected individuals or locations should be prioritized for rescue and relief distribution after a flood event [1].

Mobile crowdsensing has also been explored as a method for collecting disaster-related information from affected populations. Cicek and Kantarci conducted a comprehensive review on mobile crowdsensing applications in disaster management, highlighting how smartphones and IoT devices can collect real-time information such as victim locations and environmental conditions. While this approach improves situational awareness during emergencies, the collected data is primarily used for reporting and monitoring purposes. The system does not include intelligent mechanisms to automatically prioritize rescue operations or guide ration distribution based on the severity of the situation [2].

Geographic Information Systems (GIS) have been widely used to support disaster response through spatial analysis and mapping. Parajuli et al. developed a GIS-based evacuation route planning system that identifies safe routes in flood-prone regions. The system analyzes geographic and environmental conditions to guide evacuation planning and improve safety. Although the approach effectively supports evacuation planning, it does not incorporate machine learning techniques to predict rescue priorities or support post-flood relief management activities [3].

Machine learning models have also been used to analyze flood risk and vulnerability. Wu et al. proposed a flood disaster risk assessment framework using a Random Forest model combined with GIS-based spatial analysis. The model evaluates several environmental indicators to identify flood-prone regions and generate risk maps. While this system improves flood risk prediction and spatial analysis, it focuses mainly on identifying potential flood hazards rather than supporting rescue coordination and relief distribution during the post-flood phase [4].

Optimization techniques have been explored to improve relief logistics and resource distribution after disasters. Sahoo introduced a two-stage optimization model designed to manage relief logistics under uncertain conditions. The model determines optimal distribution strategies for emergency supplies such as food and medicine. Although this method enhances resource allocation efficiency, it relies heavily on mathematical optimization and does not

utilize machine learning models to predict which affected areas require urgent rescue and relief support [5].

Fairness-based resource allocation strategies have also been investigated in disaster management research. Long et al. proposed a dynamic resource allocation model that considers fairness and equity when distributing disaster relief resources. The model attempts to balance efficiency and fairness in humanitarian assistance. However, the framework focuses primarily on distribution strategies and assumes that demand levels are already known, without providing an intelligent mechanism to identify and prioritize affected populations in real time [6].

Spatial visualization techniques such as heatmaps have been used to analyze disaster impact zones. Wang et al. developed a GIS-based heatmap system that visualizes population density and disaster impact levels across geographic regions. The heatmap allows decision-makers to quickly identify highly affected areas and allocate resources accordingly. Despite its usefulness for visualization, the system does not incorporate predictive models that can dynamically determine rescue priorities [7].

Crowdsourced disaster mapping platforms have also been introduced to support emergency response operations. Meier discussed the use of digital humanitarian platforms that allow volunteers to map disaster impacts using satellite images and crowdsourced reports. These platforms improve disaster awareness and help authorities understand the spatial distribution of affected populations. However, the systems mainly support data collection and visualization and lack automated intelligence to predict rescue priority levels or guide ration distribution decisions [8].

Recent studies have also explored integrating machine learning with geospatial analysis for disaster vulnerability assessment. Dufitimana et al. proposed a machine learning framework that combines environmental and socioeconomic data to map vulnerability to natural hazards. The system improves the understanding of which communities are most at risk during disasters. However, the framework mainly supports long-term planning and vulnerability assessment rather than immediate rescue prioritization during disaster response [9].

Explainable artificial intelligence techniques have also been introduced to improve transparency in disaster prediction models. Derman and Mete developed a GeoAI-based flood risk analysis framework that uses explainable machine learning methods to interpret flood prediction results. This approach enhances trust and interpretability of machine learning models in disaster management. Nevertheless, the system mainly focuses on flood risk prediction and does not address post-flood rescue coordination or relief management [10].

Disaster logistics research has also investigated emergency material distribution frameworks. Li et al. proposed a two-stage emergency material distribution model designed for urban flood disasters. The framework considers urgency levels, travel time, and supply constraints when distributing relief materials. Although the approach improves logistics efficiency, it does not include intelligent classification models that automatically determine rescue priorities based on integrated disaster data [11].

Finally, research on disaster assembly area identification has demonstrated the value of geospatial analysis for emergency planning. Karakaya et al. developed a machine learning and GIS-based framework to identify safe assembly locations in flood-prone regions. The system helps authorities determine suitable locations for evacuations during disasters. However, the

approach focuses mainly on evacuation planning rather than prioritizing rescue operations and relief distribution during the post-flood recovery phase [12].

1.3 Research Gap

Although much research has been conducted on flood detection and prevention systems with minimal attention given to post-flood rescue operations and efficient ration distribution, creating a significant gap in end-to-end disaster management solutions. In particular, there is a lack of integrated systems that support real-time decision-making for rescue coordination and efficient distribution of relief resources. Current solutions do not adequately address the need to identify safe zones, locate refugee camps, and dynamically assess their conditions based on multiple factors such as population size, resource availability, and health risks.

Furthermore, existing systems are insufficient in prioritizing affected locations and resources. They typically lack intelligent mechanisms to rank refugee camps based on urgency, resulting in inefficient allocation of aid and delays in response. Additionally, there is minimal support for item-level prioritization, where essential supplies such as food, water, medicine, and sanitary items are distributed based on specific needs. The absence of integration between flood data, health data (such as disease outbreaks), and logistical considerations (such as safe route planning) further limits the effectiveness of current approaches. As a result, disaster response efforts remain largely manual, fragmented, and reactive rather than proactive and data-driven.

This research addresses these limitations by proposing an integrated, intelligent system that extends beyond traditional flood monitoring to support post-disaster response and recovery. The novelty of this approach lies in combining machine learning-based camp prioritization with rule-based item prioritization, while also incorporating disease detection insights and safe route optimization. Unlike existing systems, this solution not only identifies where assistance is needed but also determines what type of aid should be prioritized and how it should be delivered efficiently. By leveraging real-time data and advanced analytics, the proposed system provides a comprehensive, evidence-based framework for improving coordination, reducing response time, and enhancing the overall effectiveness of disaster management operations.

2. OBJECTIVE

2.1 Main Objective

To develop and evaluate an intelligent post-flood management system that utilizes real-time data and a machine learning model to accurately prioritize refugee camps and required relief resources, and to optimize delivery routes, with the aim of improving relief distribution efficiency and reducing response time by at least 30% within simulated disaster scenarios during the project period.

2.2 Specific Objectives

- To identify and map refugee camps located within safe zones by integrating flood risk data and geospatial analysis techniques.

- To collect and analyze real-time camp data, including population, resource availability, and health conditions, to support informed decision-making.

- To develop and implement a machine learning model to classify and prioritize refugee camps based on urgency using multiple contextual factors.

- To design a rule-based mechanism for prioritizing relief items (e.g., food, water, medicine, sanitary items) based on camp-specific needs and disease risk indicators.

- To determine the safest and shortest routes for relief distribution by applying pathfinding algorithms using real-time road and flood condition data.

- To visualize camp locations, priority levels, required resources, and optimal delivery routes through an interactive disaster management map.

3. METHODOLOGY

This study adopts a Design Science Research (DSR) approach combined with an experimental evaluation method to develop and assess an intelligent post-flood management system. The Design Science approach is appropriate as the research focuses on designing and implementing a practical solution to address real-world challenges in disaster response and recovery. The system is iteratively developed, tested, and refined to ensure its effectiveness in dynamic flood scenarios. In addition, an experimental approach is employed to evaluate the performance of the system under simulated conditions, enabling a comparison between the proposed intelligent solution and traditional response methods in terms of efficiency and response time.

The proposed system integrates machine learning and algorithmic techniques to support data-driven decision-making in post-flood scenarios. A supervised machine learning model, specifically XGBoost (Extreme Gradient Boosting), is utilized to classify and prioritize refugee camps based on multiple contextual factors such as population size, availability of resources, and health-related risks. To complement this, a rule-based mechanism is implemented to determine the priority of relief items, including food, water, medicine, and sanitary supplies, based on camp-specific needs and conditions. Geospatial analysis techniques are used to identify safe zones and locate refugee camps using flood data, while graph-based pathfinding algorithms such as Dijkstra or A* are applied to determine the safest and shortest routes for distributing aid, considering road conditions and flood-affected areas.

The system is developed using a modern web-based technology stack to ensure scalability, usability, and real-time performance. The frontend is implemented using React.js, which enables the creation of an interactive user interface for visualizing maps, camp locations, priority levels, and resource requirements. The backend is developed using Node.js to handle server-side logic, real-time data processing, and API integration efficiently. MySQL is used as the database management system to store structured data such as camp details, resource availability, and prediction results due to its reliability and strong relational data support. Additionally, mapping technologies such as Leaflet or Google Maps API are integrated for geospatial visualization, while the machine learning model is developed using Python-based libraries such as Scikit-learn or XGBoost to ensure high accuracy and performance.

Data for this research is collected from a combination of simulated datasets, publicly available flood-related data sources, and generated data representing refugee camp conditions. The dataset includes attributes such as population size, resource availability, environmental conditions, and health indicators. Ethical considerations are carefully addressed by ensuring that no personally identifiable or sensitive information is used in the system. All data is anonymized and utilized solely for research and simulation purposes. Furthermore, the system is designed to ensure fairness and avoid bias in decision-making, particularly in prioritizing camps and allocating resources during disaster scenarios.

The validation of the proposed system is conducted through experimental simulations to assess its effectiveness and reliability. The performance of the machine learning model is evaluated using standard metrics such as accuracy, precision, recall, and F1-score. In addition, system-level performance is measured based on improvements in response time, efficiency of resource allocation, and accuracy of prioritization compared to baseline methods. The effectiveness of route optimization is evaluated by analyzing travel distance and time reductions achieved through the selected paths. The system is tested across multiple simulated flood scenarios to

ensure robustness, scalability, and practical applicability in real-world disaster management contexts.

4. HIGH-LEVEL SYSTEM ARCHITECTURE

4.1 Overview of the Proposed Solution and its Components

The proposed system is an intelligent post-flood management platform designed to enhance disaster response and recovery through data-driven decision-making. The architecture consists of multiple interconnected components, including flood map generation, drainage monitoring and prediction, disease detection, and a relief prioritization and distribution module. These components work together to transform raw environmental and situational data into actionable insights. The system identifies safe zones, detects refugee camps, analyzes camp conditions, predicts priority levels using a machine learning model, determines required relief items, and generates optimal delivery routes. The integration of these components enables efficient coordination and timely resource allocation during post-flood operations.

4.2 System Diagram of Components

The system architecture begins with data collection from IoT sensors and external data sources, which provide real-time information such as water levels and environmental conditions. This data is processed by the flood monitoring module to generate flood maps and analyze water level variations. Based on this analysis, a safe zone identification module determines areas that are secure for relief operations. Subsequently, the system detects refugee camps located within these safe zones, followed by a camp data collection process that gathers relevant information such as population size, resource availability, and health conditions. The collected data is then processed in parallel by two components: a disease detection model, which identifies potential health risks, and a machine learning-based priority model (XGBoost), which classifies camps based on urgency. The outputs from these components are integrated to perform rule-based item prioritization, determining the most critical relief supplies required for each camp. Following this, a route optimization module utilizes algorithms such as Dijkstra or A* to identify the safest and shortest paths for delivering aid. Finally, all processed information, including camp locations, priority levels, required resources, and optimized routes, is presented through a visualization dashboard, enabling efficient monitoring and decision-making during post-flood disaster response.

4.3 Individual Component and Its Parts

The primary focus of this research is the **Relief Prioritization and Distribution Component**, which plays a critical role in post-flood response. This component consists of several sub-parts, including safe zone-based camp identification, camp data collection, machine learning-based camp prioritization, rule-based item prioritization, and route optimization. The machine learning model analyzes multiple features such as population size, resource availability, and health risks to classify camps based on urgency. The rule-based mechanism further determines

the priority of specific relief items required at each camp. Finally, route optimization ensures that resources are delivered through the safest and most efficient paths.

4.4 Component Interactions

The system operates through strong interactions between its components. The flood monitoring and drainage modules provide real-time data that is used to identify safe zones. These safe zones are then used by the refugee camp detection module to locate camps suitable for relief operations. The camp data collection module gathers relevant information, which is shared with both the disease detection component and the machine learning model. The outputs from these components are combined to determine camp priority and required resources. Subsequently, the route optimization module uses this information along with map data to generate safe delivery paths. Finally, all processed data is integrated into the visualization dashboard, allowing users to monitor and manage disaster response effectively.

4.5 Data Flow Explanation

The data flow in the system begins with the collection of real-time data from IoT sensors and external sources, including flood levels, environmental conditions, and camp-related information. This data is processed to generate flood maps and identify safe zones. Once refugee camps are detected within these zones, relevant data such as population, available resources, and health conditions are collected. This data is then fed into the machine learning model to predict camp priority levels, while disease-related data influences medicine prioritization. The outputs are passed to the item prioritization module, which determines the urgency of required supplies. Finally, the route optimization module calculates the safest and shortest paths for distribution, and all results are visualized on the system dashboard.

4.6 Scalability and Security Considerations

The system is designed with scalability in mind to handle increasing data volumes and multiple disaster scenarios. The use of modular architecture allows individual components to be independently updated or scaled as needed. Technologies such as Node.js support asynchronous processing, enabling efficient handling of real-time data streams. Additionally, cloud-based deployment can be utilized to ensure system availability and performance during high-demand situations.

From a security perspective, the system ensures data protection through secure APIs, authentication mechanisms, and role-based access control. Sensitive data is anonymized to protect user privacy, and secure communication protocols are implemented to prevent unauthorized access. These measures ensure that the system remains reliable, secure, and trustworthy for disaster management operations.

5. USER REQUIREMENTS

5.1 Stakeholder Identification

The proposed system involves multiple stakeholders who either interact directly with the system or benefit from its outcomes:

- **Government Authorities (Disaster Management Center)**
  Responsible for monitoring flood situations, making decisions, and coordinating relief operations.
- **Emergency Response Teams / Non-Governmental Organizations (NGOs)**
  Utilize the system to identify priority areas and efficiently distribute relief resources.
- **Refugee Camp Coordinators**
  Provide real-time data regarding camp conditions, resource availability, and health status.
- **Healthcare Providers**
  Monitor potential disease outbreaks and support medical resource allocation.
- **System Administrators**
  Manage system operations, maintain data integrity, and control user access.
- **Affected Communities (Indirect Stakeholders)**
  Benefit from improved response time and efficient distribution of relief resources.

5.2 Functional Requirements

The proposed system is expected to provide a set of core functionalities to support post-flood disaster management. The system shall collect real-time data from IoT sensors and external sources, including environmental conditions and flood levels, to ensure up-to-date situational awareness. Based on this data, the system shall generate flood maps to identify affected areas and determine safe zones for relief operations. It shall further detect and map refugee camps located within these safe zones.

The system shall collect and store relevant camp-related data, such as population size, available resources, and health conditions, to support data-driven decision-making. Additionally, the system shall analyze health-related data to identify potential disease risks within camps. A supervised machine learning model, specifically XGBoost, shall be utilized to classify and prioritize refugee camps based on urgency, considering multiple contextual factors.

To complement this, the system shall implement a rule-based mechanism to determine the priority of relief items, including food, water, medicine, and sanitary supplies, based on the specific needs of each camp. Furthermore, the system shall calculate optimal delivery routes using graph-based pathfinding algorithms such as Dijkstra or A*, ensuring safe and efficient distribution of resources. All processed information, including camp locations, priority levels, required resources, and delivery routes, shall be visualized through an interactive map-based dashboard. The system shall also provide real-time updates to support timely decision-making and shall incorporate user authentication and role-based access control mechanisms to ensure secure system usage.

5.3 Non-Functional Requirements

Security:
The system must ensure that sensitive information, such as victim location data, is protected through secure authentication, access control mechanisms, and encrypted communication. Only authorized users such as disaster management officials and rescue teams should be able to access system data.

Performance:
The system should process disaster-related inputs and generate rescue priority predictions within a short time frame to support rapid decision-making during emergency situations. Heatmap visualizations and dashboard updates should be generated quickly to provide real-time insights.

Usability:
The system interface and dashboard should be user-friendly and easy to interpret so that rescue teams and disaster management authorities can quickly understand priority zones and coordinate rescue operations without technical difficulties.

Scalability:
The system should be capable of handling increasing volumes of disaster-related data and user requests during large-scale flood events without significant performance degradation.

Reliability:
The system must provide consistent and accurate results during disaster response operations and ensure that rescue priority predictions and heatmap visualizations remain available when needed.

Maintainability:
The system should be designed in a modular manner so that future updates, integration with additional disaster monitoring modules, or improvements to the machine learning model can be implemented easily.

6. COMMERCIALIZATION PLAN

6.1 Target Market and Customer Persona

The primary target market for the proposed intelligent post-flood rescue and ration management system includes **government disaster management authorities, emergency response organizations, and humanitarian agencies** responsible for managing disaster response operations. These organizations require reliable decision-support tools to improve rescue coordination and relief distribution during flood emergencies. In addition, **local government institutions, municipal councils, and international humanitarian organizations** involved in disaster management can benefit from the system.

The main customer persona includes **disaster management officers and emergency coordinators** who are responsible for planning rescue operations and allocating resources during disaster situations. These users require a system that provides accurate information about affected areas, rescue priority levels, and relief demand. Another important user group is **rescue team leaders and field officers**, who rely on visual dashboards and geospatial maps to quickly identify critical locations and coordinate rescue activities in the field.

6.2 Value Proposition

The proposed system offers a comprehensive and intelligent solution for post-flood disaster management by integrating machine learning, geospatial analysis, and real-time data processing into a unified platform. Unlike traditional systems that primarily focus on flood prediction and monitoring, this solution addresses the critical gap in post-disaster response by enabling efficient coordination of relief operations.

The system delivers value by improving the accuracy and speed of decision-making in disaster scenarios. It enables authorities to prioritize refugee camps based on urgency, allocate resources based on actual needs, and identify the safest and most efficient delivery routes. This leads to reduced response time, optimized resource utilization, and improved outcomes for affected populations. Furthermore, the system enhances transparency, coordination, and accountability among stakeholders, thereby contributing to more resilient and sustainable disaster management practices.

6.3 Revenue Model

The proposed system can adopt multiple revenue generation strategies to ensure sustainability and scalability. One primary approach is a **Software-as-a-Service (SaaS)** model, where government agencies, disaster management authorities, and humanitarian organizations subscribe to the platform on a periodic basis.

Additionally, a **licensing model** can be implemented, allowing organizations to deploy the system within their own infrastructure. The system may also offer **customization and integration services** tailored to specific regional or organizational requirements, generating additional revenue.

Furthermore, partnerships with international organizations, non-governmental organizations (NGOs), and development agencies can provide funding and support for deployment in disaster-prone regions. Grants and public sector funding may also be leveraged, given the system's alignment with disaster resilience and sustainable development initiatives.

6.4 Cost Estimation

The cost structure of the proposed system includes both development and operational expenses. Development costs involve software design, machine learning model development, system integration, and user interface design. These costs also include the use of development tools, cloud services, and testing environments.

Operational costs include cloud hosting, data storage, system maintenance, and technical support. Additional costs may arise from acquiring or integrating real-time data sources such as IoT sensors and third-party APIs. Human resource costs, including developers, data scientists, and system administrators, also contribute significantly to the overall expenditure.

Despite these costs, the system is designed to be scalable and cost-efficient in the long term, particularly when deployed using cloud-based infrastructure and modular architecture.

6.5 Pricing Strategy

The pricing strategy for the proposed system is designed to ensure accessibility while maintaining financial sustainability. A **tiered pricing model** can be adopted, where different packages are offered based on the scale of usage, number of users, and level of functionality required.

For example, a basic package may provide essential features such as flood mapping and camp identification, while advanced packages may include machine learning-based prioritization, route optimization, and real-time analytics. Pricing may also vary based on geographic coverage and system customization requirements.

For government and humanitarian organizations, flexible pricing or subsidized models can be considered to encourage adoption, especially in developing regions. In addition, pilot deployments and trial versions may be offered to demonstrate system effectiveness before full-scale implementation.

6.6 Competitive Advantage

The proposed system offers a significant competitive advantage by providing an integrated and intelligent approach to post-flood disaster management. Unlike existing solutions that focus on isolated functionalities such as flood prediction or resource distribution, this system combines multiple capabilities into a single platform.

Key differentiators include the use of machine learning for dynamic camp prioritization, rule-based mechanisms for item prioritization, and advanced route optimization for efficient logistics. The integration of real-time data, health risk analysis, and geospatial visualization further enhances its effectiveness.

Moreover, the system's ability to support end-to-end decision-making—from identifying affected areas to delivering aid—positions it as a comprehensive solution in the disaster management domain. Its scalability, adaptability, and data-driven approach provide a strong advantage over traditional and fragmented systems.

6.7 Intellectual Property Considerations

The proposed system involves several innovative components that may be eligible for intellectual property protection. The integration of machine learning-based prioritization with rule-based resource allocation and route optimization represents a novel approach in post-disaster management.

The software system, including its architecture, algorithms, and user interface design, can be protected under copyright laws. Additionally, specific components such as the prioritization model and decision-support mechanisms may be considered for patent protection, subject to novelty and legal requirements.

It is also important to ensure compliance with data protection regulations and ethical standards, particularly when handling sensitive or real-time data. Any third-party tools, libraries, or datasets used in the system must adhere to their respective licensing agreements. Proper documentation and ownership of developed components should be maintained to safeguard intellectual property rights.

7. BUDGET AND JUSTIFICATION

The development of the intelligent post-flood rescue and ration management system requires resources related to hardware, software, cloud infrastructure, and human effort. The estimated budget below outlines the major cost categories required to design, develop, test, and deploy the proposed system. All costs are estimated in **Sri Lankan Rupees (LKR)**.

7.1 Hardware Costs

Item	Estimated Cost (LKR)	Justification
Development Laptop / Workstation	250,000	Required for software development, machine learning model training, and system testing.
External Storage / Backup Drive	30,000	Used for storing datasets, backups, and disaster-related spatial data securely.
Mobile Device for Testing	70,000	Used to test victim reporting interfaces and field data collection features.

Table 1 - Hardware Costs

Total Hardware Cost: 350,000 LKR

Hardware is necessary to support development, data processing, testing, and simulation of the proposed system during the research and implementation phases.

7.2 Software Licenses

Item	Estimated Cost (LKR)	Justification
Development Tools and IDEs	0	Open-source tools such as Python, VS Code, and machine learning libraries will be used, reducing licensing costs.
GIS	50,000	Optional licenses for advanced geospatial visualization tools may be required for system development and analysis.

Table 2 - Software Licenses

Total Software Cost: 50,000 LKR

Most tools used in the project will be **open-source**, minimizing software licensing costs while still supporting machine learning and geospatial analysis.

7.3 Cloud Services

Item	Estimated Cost (LKR)	Justification
Cloud Server Hosting	60,000	Cloud infrastructure required for deploying the system, hosting databases, and supporting web or mobile access.
Data Storage Services	40,000	Storage required for disaster data, victim location information, and system outputs.
API and Map Services	30,000	Integration of map-based services and APIs for heatmap visualization and geographic analysis.

Table 3 - Cloud Services

Total Cloud Service Cost: 130,000 LKR

Cloud infrastructure allows the system to scale during disaster events and supports remote access for authorities and rescue teams.

7.4 Human Effort Cost

Item	Estimated Cost (LKR)	Justification
Software Development Effort	300,000	Time and effort required to design, develop, and implement the system modules.
Machine Learning Model Development	150,000	Effort required for data processing, model training, and evaluation of the Random Forest algorithm.
System Testing and Evaluation	100,000	Effort required for system testing, debugging, and validation of results.

Table 4 - Human Effort Cost

Total Human Effort Cost: 550,000 LKR

Human effort represents the most significant investment in developing the system, including system design, coding, machine learning implementation, and evaluation.

7.5 Total Estimated Budget

Category	Cost (LKR)
Hardware	350,000
Software	50,000
Cloud Services	130,000
Human Effort	550,000

Table 5 - Total Estimated Budget

Total Estimated Project Cost: 1,080,000 LKR

Budget Justification

The proposed budget supports the development of a scalable and reliable disaster response system. Hardware resources ensure efficient development and testing, while cloud services enable remote access and scalable deployment during disaster situations. The use of open-source software significantly reduces licensing costs. The majority of the budget is allocated to human effort because the project requires significant time for system design, machine learning model development, and evaluation. This investment is justified by the potential benefits of the system in improving post-flood rescue coordination and relief distribution efficiency.

8. WORK BREAKDOWN STRUCTURE (WBS)

8.1 Task Breakdown per Member

The project is divided among four team members, where each member is responsible for developing a specific component of the overall **post-flood disaster response and recovery management system**.

• **Member 1 – Flood Risk Mapping Component**
Responsible for analyzing environmental and geographical data to identify flood-prone areas and generate flood risk levels. This component produces flood risk indicators that are later used by other modules.

• **Member 2 – Drain Water Level Monitoring Component**
Responsible for collecting and analyzing drainage and water level information from monitoring sources. This component generates real-time flood severity alerts and water level indicators.

• **Member 3 – Post-Flood Disease Detection Component**
Responsible for analyzing health-related data to identify potential disease risks in flood-affected areas. This component produces health risk alerts that support disaster management planning.

• **Member 4 – Post-Flood Relief Prioritization and Distribution Component**
Responsible for analyzing post-flood conditions to prioritize refugee camps, determine required relief resources, and optimize distribution processes. This component utilizes machine learning techniques to classify camps based on urgency using factors such as population size, resource availability, and health risks. It also applies a rule-based mechanism to prioritize essential relief items, including food, water, medicine, and sanitary supplies. Additionally, it incorporates route optimization algorithms to identify the safest and most efficient paths for delivering aid. This component produces prioritized camp lists, recommended resource allocations, and optimized delivery routes to support effective disaster response.

8.2 Timeline Alignment

The development and implementation of the proposed intelligent post-flood management system are structured across multiple phases to ensure systematic progress and effective integration of all components. The timeline is aligned with the overall research objectives and divided into sequential stages, allowing each team member to contribute to their respective components while maintaining coordination across the system.

During the **initial phase**, requirements analysis and literature review are conducted to establish a strong theoretical and technical foundation for the system. At this stage, all team members collaboratively define system requirements, identify relevant data sources, and design the overall system architecture.

The **design and data preparation phase** focuses on developing individual component designs and preparing datasets required for implementation. Member 1 develops the flood risk mapping models, Member 2 focuses on water level monitoring mechanisms, Member 3 prepares datasets and models for disease detection, and Member 4 designs the framework for relief prioritization and distribution, including defining features for machine learning and rule-based decision-making.

In the **development phase**, each component is implemented independently. Member 4 specifically develops the machine learning model for camp prioritization, implements the rule-based system for resource allocation, and integrates route optimization algorithms. Parallel development ensures efficient use of time while maintaining component-level focus.

The **integration phase** involves combining all individual components into a unified system. Outputs from flood mapping, water monitoring, and disease detection modules are integrated into the relief prioritization component to enable comprehensive decision-making. During this phase, data flow between components is tested and refined.

The **testing and evaluation phase** includes system validation using simulated flood scenarios. Performance metrics such as prioritization accuracy, response time, and route efficiency are evaluated. Member 4 contributes by assessing the effectiveness of the prioritization model and the efficiency of resource allocation and route optimization.

Finally, the **deployment and documentation phase** focuses on system refinement, report writing, and preparation for presentation. All members collaborate to finalize the system, document findings, and demonstrate the effectiveness of the proposed solution.

This structured timeline ensures that all components are developed cohesively while allowing each team member to focus on their specific responsibilities. It also supports iterative improvement and effective integration, ultimately contributing to the successful completion of the research project.

8.3 Balanced Workload Distribution

The workload of the proposed system development is distributed among team members in a balanced and structured manner to ensure efficiency, accountability, and effective collaboration. Each team member is assigned a specific component based on the overall system architecture, allowing for parallel development while maintaining clear responsibilities.

Member 1 is responsible for the Flood Risk Mapping Component, which involves analyzing environmental and geographical data to generate flood risk indicators. Member 2 handles the Drain Water Level Monitoring Component, focusing on collecting and processing real-time water level data to produce flood severity alerts. Member 3 is assigned the Post-Flood Disease Detection Component, which analyzes health-related data to identify potential disease risks in affected areas.

Member 4 is responsible for the Relief Prioritization and Distribution Component, which plays a critical role in post-disaster response. This includes developing the machine learning model

for camp prioritization, implementing rule-based mechanisms for resource allocation, and integrating route optimization algorithms to ensure efficient delivery of aid.

The distribution of tasks ensures that each component has a comparable level of complexity and technical depth, incorporating aspects such as data processing, algorithm design, and system integration. Additionally, all team members collaboratively contribute to shared activities, including system design, data preparation, integration, testing, and documentation.

Regular communication and coordination among team members are maintained to ensure consistency across components and to address interdependencies effectively. This balanced allocation of responsibilities supports timely completion of the project while promoting teamwork and equal contribution from all members.

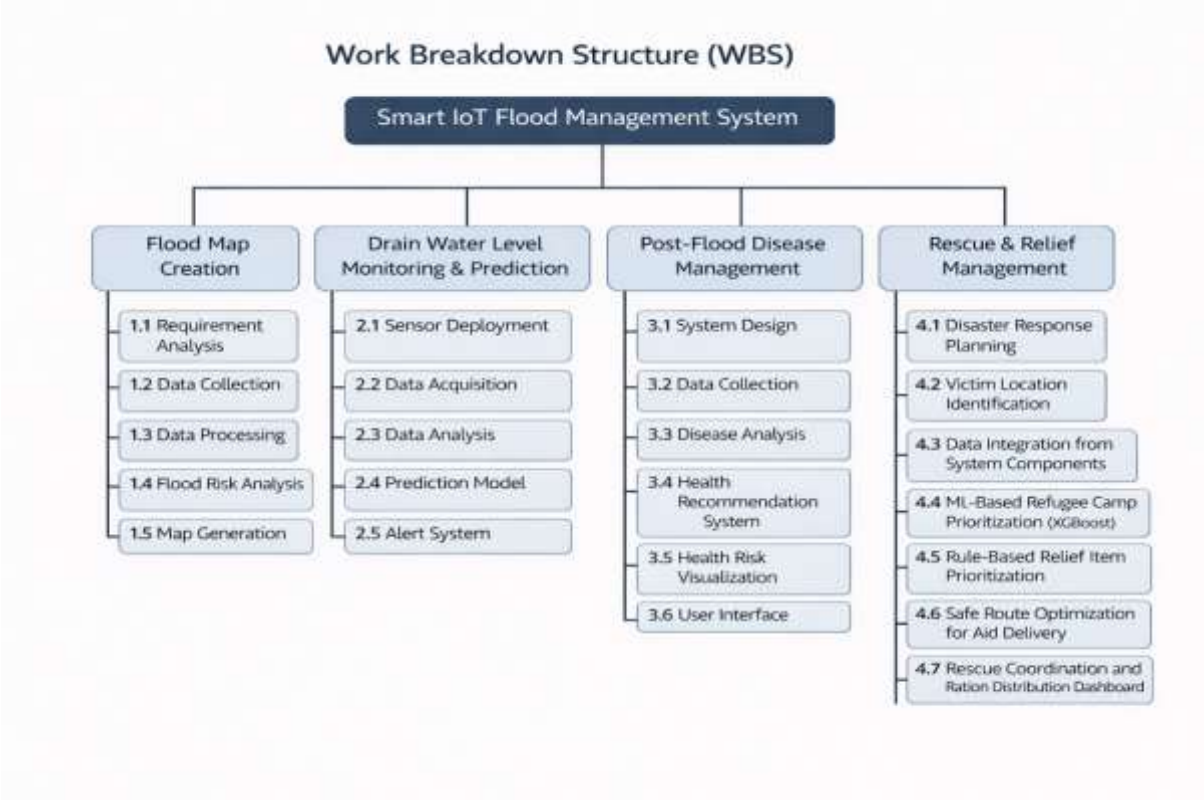


Figure 1 - Work Breakdown Structure

9. GANTT CHART

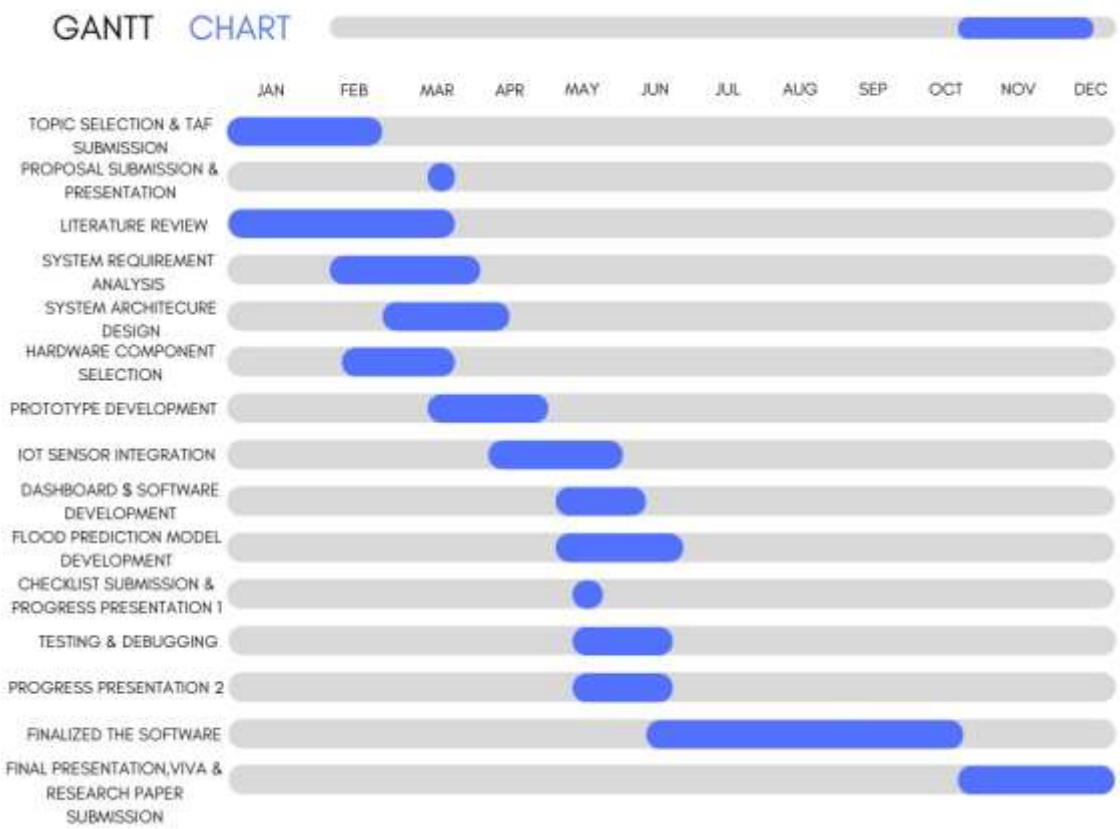


Figure 2 - GANTT CHART

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11.APPENDICES

The appendices section includes additional materials that support the development and evaluation of the proposed **intelligent post-flood rescue and ration distribution management system**. These materials provide supplementary information such as data collection tools, system design diagrams, interface designs, and ethical considerations used in the research process.

11.1 Survey Instruments

Survey instruments are used to collect information from disaster management personnel, rescue teams, and affected communities to understand challenges in rescue coordination and relief distribution after flood disasters. The survey helps identify user needs, system requirements, and practical limitations faced during emergency response operations.

Examples of survey questions include:

- What are the major challenges faced when identifying affected individuals during flood disasters?
- How do rescue teams currently prioritize rescue operations in flood-affected areas?
- What types of information are most useful for planning rescue and relief operations?
- How frequently do delays occur in distributing relief supplies during flood emergencies?
- What features would be most useful in a digital disaster response support system?

These responses will help refine the system design and ensure that the proposed solution addresses real-world operational needs.

Appendices 1 - Survey Instruments

11.2 Detailed Diagrams

Detailed diagrams illustrate the internal structure and workflow of the proposed system. These diagrams help explain how different modules interact and how data flows through the system.

- **System Architecture Diagram**

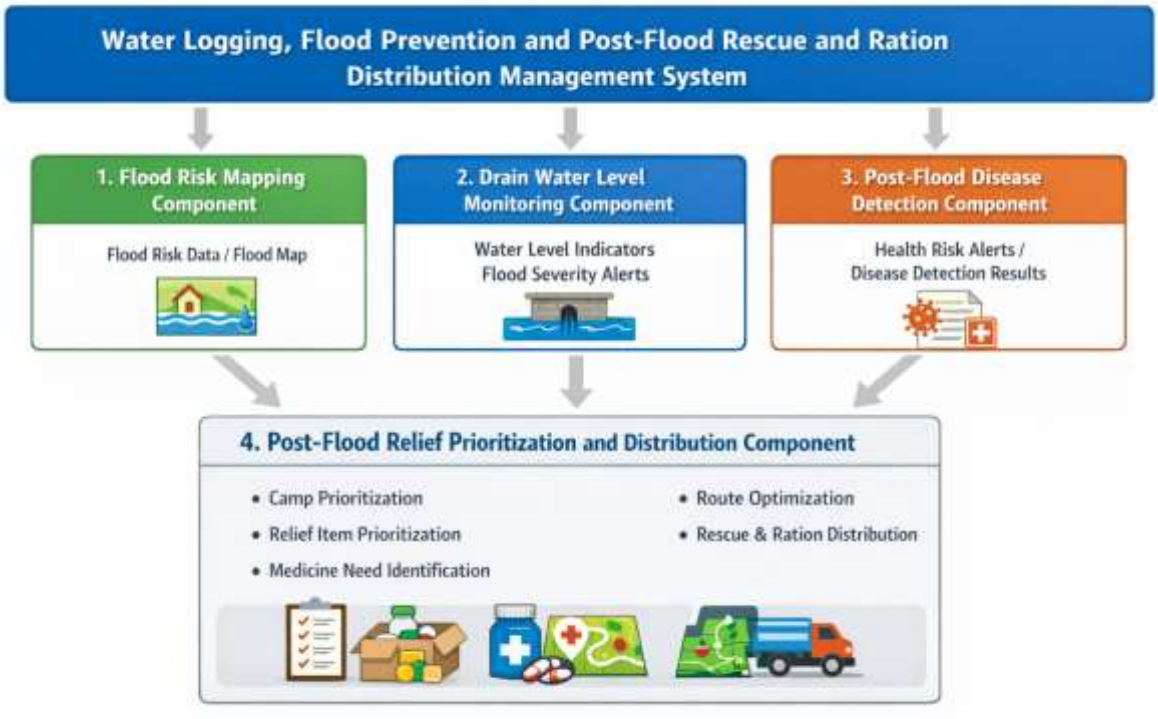


Figure 3 - System Architecture Diagram

- **Machine Learning Workflow**

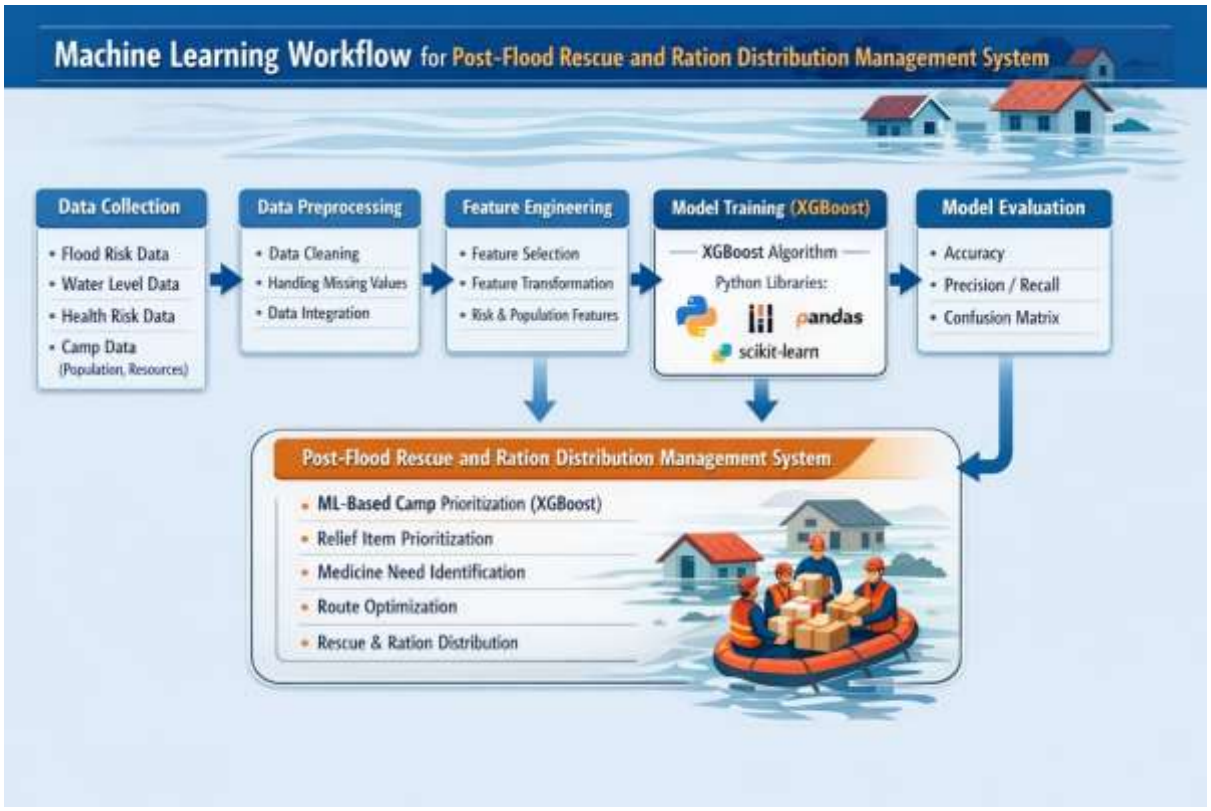


Figure 4 - Machine Learning Workflow Diagram

Appendices 2 - Detailed Diagrams

11.3 Risk Analysis

Risk analysis identifies potential risks that may affect the development or deployment of the proposed system and provides mitigation strategies to reduce these risks.

Examples of possible risks include:

- **Data availability issues:** Limited or incomplete disaster data may affect machine learning predictions.
- **System integration challenges:** Integrating outputs from multiple system components may create technical complexities.
- **Infrastructure limitations:** Poor internet connectivity in disaster-affected areas may limit real-time data transmission.
- **Model accuracy concerns:** Machine learning predictions may require continuous improvement through additional data and testing.
- **User adoption challenges:** Rescue teams may require training to effectively use the system.

Appropriate mitigation strategies such as data validation, modular system design, offline data collection mechanisms, and user training programs will be considered.

Appendices 3 - Risk Analysis

11.4 Ethical Documents

Ethical considerations are essential in research involving human-related data and disaster response information. The proposed system handles sensitive data such as affected individuals' locations and emergency reports; therefore, appropriate ethical practices must be followed.

Examples of ethical considerations include:

- **Data privacy protection:** Personal information will be anonymized and stored securely.
- **Informed consent:** Participants involved in surveys or interviews will be informed about the purpose of the research.
- **Responsible data usage:** Disaster-related data will be used only for research and emergency response purposes.
- **Access control:** Only authorized personnel will be allowed to access sensitive information.
- **Compliance with institutional research guidelines:** The research will follow ethical standards and policies required by the academic institution.

These ethical measures ensure that the research process respects privacy, security, and responsible data handling practices.

Appendices 4 - Ethical Documents

11.5 Plagiarism report

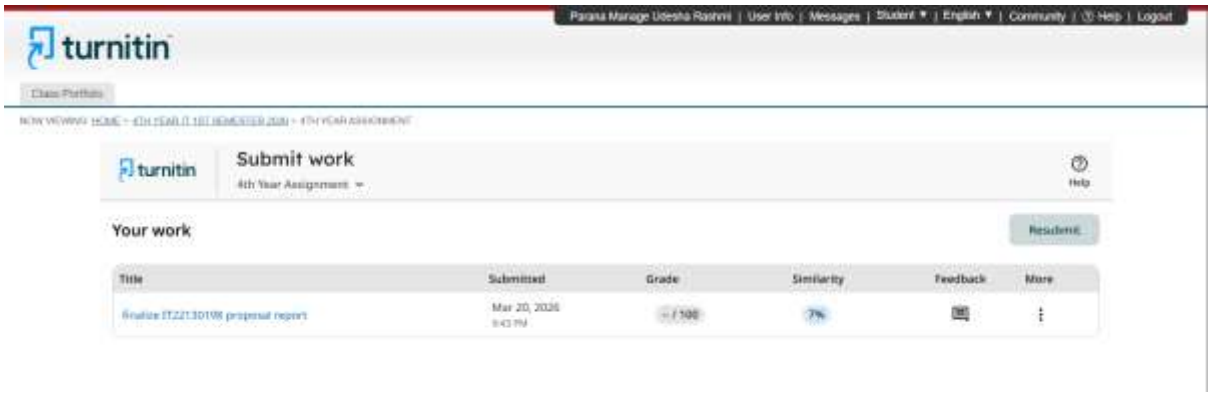


Figure 5 - Turnitin Plagiarism report

Appendices 5 - Plagiarism report